

```
Q <= '0';  
else  
if (rising_edge(B)) then  
Q <= N;  
end if;  
end if;  
end process X;  
end arch;
```

Select one:

- ☐ a. 4-bit register with Synchronous reset
- ☐ b. 4-bit register with Synchronous reset and enable
- ☐ c. 4 D latch with enable
- ☐ d. 4-bit register with Asynchronous clear
- ☐ e. 4-bit register with Asynchronous clear and enable
- ☐ f. 4-bit register without clear, reset and enable

12
answered
out of

-- Consider the state table bellow, select the right choice to make the code bellow a
described by this state diagram?

Type here to search





Photo

end XXX;

architecture arch of XXX is

begin

X: process (B, R)

begin

if (R='1') then

Q <= '0';

else

if (rising_edge(B)) then

Q <= N;

end if;

end if;

end process X;

end arch;

Select one:

- ☐ a. 4-bit register without clear, reset and enable
- ☐ b. 4-bit register with Synchronous reset and enable
- ☐ c. 4-bit register with Synchronous reset
- ☒ d. 4-bit register with Asynchronous clear



- ☐ f. 4-bit register with Asynchronous clear and enable

Question 24

Not yet answered

Marked out of 0.60

Flag question

Given the signals bellow

signal X : std_logic_vector(7 downto 0);

signal Y : unsigned(7 downto 0);

signal Z : integer;

Which of the following is a legal VHDL assignment

Select one or more:

- ☐ a. `X <= to_integer(Y);`
- ☐ b. `X <= unsigned(Y);`
- ☒ c. `Y <= to_unsigned(35, 4);`
- ☐ d. `Y (5 downto 0) <= to_unsigned(17, 6);`
- ☒ e. `Z <= to_integer(unsigned(x));`

Question 7

Not yet answered

Marked out of
0.50

🚩 Flag question

Say that $a(7 \text{ downto } 0) = '11110011'$ and $b(3 \text{ downto } 0) = '1001'$. Determine the values produced by the assignments below:

b(5 DOWNT0 2) XNOR (NOT b);

Answer: 0101

Question 8

Answer saved

Marked out of 0.60

Flag question

The V in VHDL stands for?

Select one:

- ☐ a. Very
- ☒ b. Very High Speed IC
- ☐ c. Version 5
- ☐ d. Verilog
- ☐ e. Vast

[Clear my choice](#)

☰ Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25							

Finish attempt ...

Time left 2:29:01

Activate Windows

architecture rtl of four_state_mealy_state_machine is

type state_type is (A, B, C, D, E);

attribute syn_encoding : string;

attribute syn_encoding of State_type : type is '000 001 010 011 100';

signal state, next_state : state_type;

begin

process ()

begin

if (rising_edge(clk)) then

if reset = '1' then

Next_state <= A;

else

state <= Next_state;

end if;

end if;

end process;

Type here to search



Final Exam: Introduction to x

elearning7.uokufa.edu.iq/eng/mod/quiz/attempt.php?attempt=35607&cmid=15697&page=5

🔒 ⏪ ⭐ 🌐

🔍 📞 📘 🗂️ ⬇️ ☰

Import bookmarks

output <= ;

end if;

end case;

end process;

end rtl;

– This FSM design is model

Question 12

Not yet answered

Marked out of 0.50

🚩 Flag question

Initial values for signals in VHDL are:

Select one or more:

☐ a. Are to be avoided in Testbench

☐ b. Is to be avoided in synthesizable code

☐ c. Are useful in Testbench

☐ d. Is to be avoided in non-synthesizable code

Previous page

Next page

Activate Windows
Go to Settings to activate Windows.

🪟 🔍 📄 📁 🛒 🌐 📧 📺

⬆️ 📶 🔌 🔊 🔍 🗨️ ⚙️ 04:10 ٠٨/٠٩/٢٠٢٠

The truth table below represents a FSM with a 1-bit input (x) and a 1-bit output. Suppose the machine is initialized to its starting state '10' and then processes the input values: 1,0,0,1,1,1 (1 input bit after each Clk). What are state transitions?

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

Select one:

- ☒ a. 10, 00, 00, 01, 11, 10
- ☐ b. 10, 11, 10, 00, 00, 01
- ☐ c. 10, 00, 10, 01, 00, 10
- ☐ d. 10, 11, 10, 00, 10, 00

Final Exam: Introduction to x

elearning7.uokufa.edu.iq/eng/mod/quiz/attempt.php?attempt=35607&cmid=15697&page=7

🔒 ⏪ ⭐ 🌐 🔍 📞 📧 ✂️ ⬇️ ☰

📌 Import bookmarks

🚩 Flag question

☐ c. Wrong

☒ d. failure

☒ e. Error

☐ f. Correct

17

18

19

20

21

22

23

24

25

Finish attempt ...

Time left 2:16:28

Question 16

Not yet answered

Marked out of 0.50

🚩 Flag question

In the following VHDL architecture code :
with SEL select
z <= A when '00',
B when '01',
C when '10',
D when others;
The when others covers and catches the following conditions:

Select one:

☐ a. when '1'

☐ b. when '11'

☒ c. when '00'

☐ d. when '101'

[Clear my choice](#)

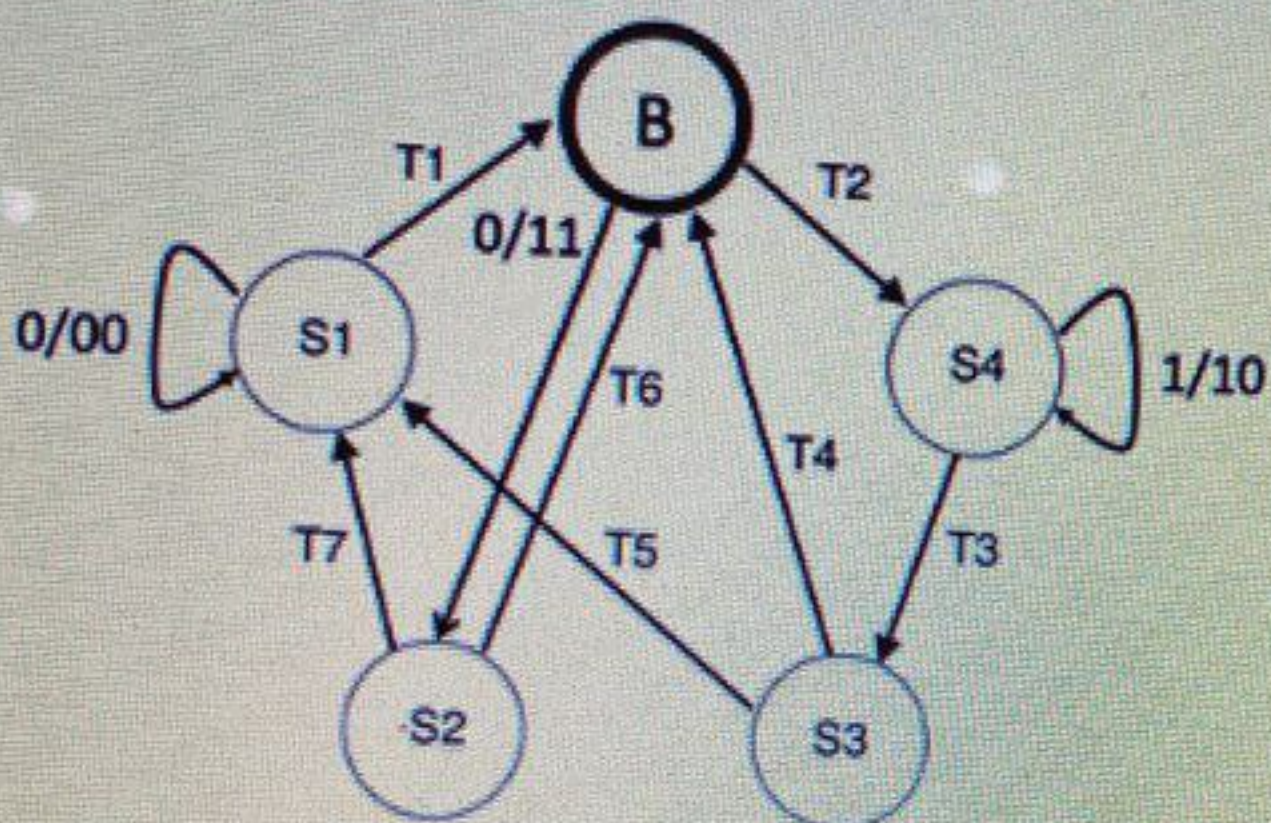
Activate Windows

Go to Settings to activate Windows.

🏠 🔍 📁 📌 🌐 📧 📞 📧 📞

04:17

E	0	A	10
E	1	B	01



T7	0/01	⊕
T2	1/00	⊕
T3	0/01	⊕
T6	1/01	⊕
T1	1/11	⊕
T5	0/10	⊕
T4	1/01	⊕

T2

Choose... ▾

T7

Choose... ▾

T6

Choose... ▾

T5

Choose... ▾

T3

Choose... ▾

T4

1/01 ▾

T1

Choose... ▾

Please answer all parts of the question.

Question 1

Not yet
answered

Marked out of
0.60

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal (more than one choice are allowed)

Select one or more:

- ☐ s4(4 downto 3) <= s3 & s2(1);
- ☐ s2 (6 DOWNT0 1) <= S2 (7 DOWNT0 2);
- ☐ s2(3 to 0) <= s4(5 to 2);
- ☐ s1 <= '11';

Question 2

Answer saved

Marked out of
0.60

To design a counter that counts the even numbers from 0 to 20, how many D-flip flops we will need ?

Select one:

Quiz navigation

1	2	3
7	8	9
13	14	15
19	20	21
25		

Finish attention

Time left 2

Question 21
Not yet answered
Marked out of 0.50
Flag question

The following VHDL statements deduce a flip-flop :

Select one or more:

- ☐ a. if (clk'event and clk='1') then q<= d;
- ☐ b. if (rising_edge(clk)) then q<= d;
- ☐ c. if (clk= '1') then q<= d;
- ☐ d. q<= d when (clk='1') else q<= 0;

Question 22
Not yet answered
Marked out of 0.50
Flag question

For the FSM Mealy mode outputs of a circuit are a function of:

Select one:

- ☐ a. the present state.
- ☐ b. the next state
- ☐ c. the present state and the present inputs of the circuit.
- ☐ d. the present inputs of the circuit.
- ☐ e. the next state and the present inputs of the circuit.

Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25							

Finish attempt ...

Time left 1:55:13

Previous page

Next page

Activate Windows
Go to Settings to activate Windows.

Question 21
Not yet answered
Marked out of 0.50
Flag question

The following VHDL statements deduce a flip-flop :

Select one or more:

- ☒ a. if (clk'event and clk='1') then q<= d;
- ☒ b. if (rising_edge(clk)) then q<= d;
- ☐ c. if (clk= '1') then q<= d;
- ☐ d. q<= d when (clk='1') else q<= 0;

Question 22
Not yet answered
Marked out of 0.50
Flag question

For the FSM Mealy mode outputs of a circuit are a function of:

Select one:

- ☐ a. the present state.
- ☐ b. the next state
- ☐ c. the present state and the present inputs of the circuit.
- ☐ d. the present inputs of the circuit.
- ☒ e. the next state and the present inputs of the circuit.

[Clear my choice](#)

Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25							

Finish attempt ...

Time left 1:52:41

Activate Windows
Go to Settings to activate Windows.

BEGIN

```
Clock_Loop: Loop  
Clk<= '0';  
wait Clock_Cycle/2;  
Clk<= '1';  
wait Clock_Cycle/2;  
end loop
```

END PROCESS;
END part4_arch;

Select one:

- ☐ a. 0000 0000
- ☐ b. 1011 0000
- ☐ c. 0010 1100
- ☐ d. 0010 0011
- ☒ e. 0011 0000
- ☐ f. 0000 1100

[Clear my choice](#)

[previous page](#)

🔍 Type here to search



end arch;

Select one:

- ☐ a. 4-bit register without clear, reset and enable
- ☐ b. 4-bit register with Asynchronous clear and enable
- ☐ c. 4 D latch with enable
- ☐ d. 4-bit register with Synchronous reset and enable
- ☐ e. 4-bit register with Synchronous reset
- ☒ f. 4-bit register with Asynchronous clear

[Clear my choice](#)

Question 6

Answer saved

Marked out of
0.50

 Flag question

Name the parts of the structure of a VHDL file.

Select one:

- ☐ a. Entity
- ☐ b. Library, Entity, Process, Components, and Architecture
- ☐ c. Library, Entity, and architecture
- ☒ d. Entity, and architecture
- ☐ e. Library, Entity, process, and architecture

[Clear my choice](#)

☐ True

☐ False

question

4

lete

out of

question

To test a 2x4 Decoder using Records, complete the following record declaration and

type test_vector is record

d : std_logic_vector(2 downto 0);

o : std_logic_vector(4 downto 0);

end record;

type test_vector_array is array (0 to 2) of test_vector;

constant test_vectors : test_vector_array := (

(, , , , ,),

Type here to search



The initial value that is assigned to Generic in component could be overwritten in the top-level entity:

Select one:

☐ True

☐ False

Describing the digital circuit in hierarchical way is useful because:

Select one or more:

- ☐ a. It enable the designer to reuse the some parts of the design
- ☐ b. Divide a complicated circuit into manageable smaller circuit.
- ☐ c. Reduce the size of the implemented circuit.
- ☐ d. Improve performance of the final design

Clear my choice

Given the the signals bellow:

SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal (more than one choice are allowed)

Select one or more:

☐ s2 (6 DOWNT0 1) <= s2 (7 DOWNT0 2);

☐ s1 <= '11';

☐ s2(3 to 0) <= s4(5 to 2);

☐ s4(4 downto 3) <= s3 & s2(1);

Question 1

Not yet answered

Marked out of
0.60

Flag question

The V in VHDL stands for?

Select one:

- ☐ a. Very
- ☐ b. Verilog
- ☐ c. Vast
- ☒ d. Very High Speed IC
- ☐ e. Version 5

[Clear my choice](#)

Question 2

Not yet answered

Marked out of
0.50

Flag question

For the FSM Mealy mode outputs of a circuit are a function of:

Select one:

- ☒ a. the present state
- ☐ b. the next state
- ☐ c. the present state and the present inputs of the circuit.
- ☐ d. the next state and the present inputs of the circuit.
- ☐ e. the present inputs of the circuit.

[Clear my choice](#)

Question 7

Answer saved

Marked out of
0.50

Flag question

In the following VHDL architecture code :

with SEL select

z <= A when '00',

B when '01',

C when '10',

D when others;

The when others covers and catches the following conditions:

Select one:

- ☐ a. when '101'
- ☒ b. when '11'
- ☐ c. when '00'
- ☐ d. when '1'

[Clear my choice](#)

Quiz navigation

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25			

Finish attempt ...

Time left 2:47:41

Question 8

Not yet answered

Marked out of
0.60

For each component in top-level entity:

Select one or more:

- ☐ a. When instantiated, it should have unique label

Name the parts of the structure of a a VHDL file?

Select one:

- ☐ a. Library, Entity, Process, Components, and Architecture
- ☒ b. Entity, and architecture
- ☐ c. Library, Entity, process, and architecture
- ☐ d. Entity
- ☐ e. Library, Entity, and architecture

[Clear my choice](#)



- ☐ f. 4-bit register with Asynchronous clear and enable

Question 24

Not yet answered

Marked out of 0.60

Flag question

Given the signals bellow

signal X : std_logic_vector(7 downto 0);

signal Y : unsigned(7 downto 0);

signal Z : integer;

Which of the following is a legal VHDL assignment

Select one or more:

- ☐ a. `X <= to_integer(Y);`
- ☐ b. `X <= unsigned(Y);`
- ☒ c. `Y <= to_unsigned(35, 4);`
- ☐ d. `Y (5 downto 0) <= to_unsigned(17, 6);`
- ☒ e. `Z <= to_integer(unsigned(x));`

Quiz navigation



Finish attempt ...

Time left 2:31:53

Question 5

Not yet answered

Marked out of 0.60

Flag question

Given the the signals bellow:

SIGNAL s1: BIT;

SIGNAL s2: BIT_VECTOR(7 DOWNT0 0);

SIGNAL s3: STD_LOGIC;

SIGNAL s4: STD_LOGIC_VECTOR(7 DOWNT0 0);

VARIABLE v1: BIT_VECTOR(7 DOWNT0 0);

Select which statement bellow are legal (more than one choice are allowed)

Select one or more:

- ☒ s2 (6 DOWNT0 1) <= S2 (7 DOWNT0 2);
- ☐ s2(3 to 0) <= s4(5 to 2);
- ☐ s1 <= '11';
- ☒ s4(4 downto 3) <= s3 & s2(1);

Question 6

Not yet answered

Marked out of 5.00

Flag question

Design a simple ALU. There are 2 inputs , A and B, and each one is stored in 16-bit register and a 3-bit opcode. The output is a 17-bit register Y. The opcode controls ALU functions as follows:

Opcode	Operation	Function
000	$Y \leftarrow A$	A

Select one:

- ☐ a. 4-bit register with Asynchronous clear and enable
- ☐ b. 4-bit register with Synchronous reset and enable
- ☐ c. 4 D latch with enable
- ☐ d. 4-bit register with Asynchronous clear
- ☐ e. 4-bit register with Synchronous reset
- ☐ f. 4-bit register without clear, reset and enable

Question 10

Not yet answered

Marked out of
0.50

Flag question

For the FSM Mealy mode outputs of a circuit are a function of:

Select one:

- ☐ a. the next state and the present inputs of the circuit.
- ☐ b. the present state and the present inputs of the circuit.
- ☐ c. the present inputs of the circuit.
- ☒ d. the next state
- ☐ e. the present state.

[Clear my choice](#)

Previous page

Next page

- ☐ j. Line 22
- ☐ k. Line 13
- ☐ l. Line 11
- ☐ m. Line 25
- ☐ n. Line 9
- ☐ o. line 7
- ☐ p. Line 4
- ☐ q. Line 18

Question 18

Not yet answered

Marked out of
0.50

Flag question

Describing the digital circuit in hierarchical way is useful because:

Select one or more:

- ☒ a. It enable the designer to reuse the some parts of the design
- ☒ b. Divide a complicated circuit into manageable smaller circuit.
- ☐ c. Reduce the size of the implemented circuit.
- ☐ d. Improve performance of the final design

[Previous page](#)[Next page](#)

Question 5

Not yet answered

Marked out of
0.60

Flag question

For each component in top-level entity:

Select one or more:

- ☐ a. When instantiated, mapping its ports is optional.
- ☐ b. When instantiated, it should have unique label
- ☐ c. It can be instantiated multiple times.
- ☐ d. It can be declared multiple times

Question 6

Answer saved

Marked out of

The following VHDL statements deduce a flip-flop :

Select one or more: